

Lab 9: Gravimetric Analysis of Chloride

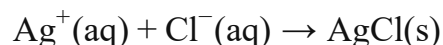
Chemistry 150 - Winter 2026

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Lab Section	2026W CHEM-150-X01AB
Date Performed	March 23, 2026
Unknown Number	379

Introduction

Gravimetric analysis is a method of determining the amount of a target species by converting it into a solid compound, filtering and drying that solid, and weighing it precisely. Because the solid has a known chemical formula, the mass of the target can be back-calculated from the mass of the precipitate.

In this experiment, the target is chloride ion (Cl^-) dissolved in an unknown sample. When silver nitrate solution (AgNO_3) is added, Ag^+ reacts with Cl^- to form insoluble silver chloride:



The extent to which this precipitation is complete is governed by the solubility product of AgCl :

$$K_{sp} = [\text{Ag}^+][\text{Cl}^-] = 1.8 \times 10^{-10}$$

If Ag^+ is kept in large excess, the equilibrium forces $[\text{Cl}^-]$ to an extremely low value, ensuring virtually all chloride is captured as precipitate.

AgCl initially forms as a colloid — particles too small to be caught by a filter. Heating the mixture and adding electrolyte (HNO_3) removes the surface charges that keep the particles apart, causing them to clump into larger particles that can be filtered. AgCl is sensitive to light: exposure reduces Ag^+ to grey or black Ag metal, so the mixture is stored in a dark cupboard during aging.

Objective

Determine the mass percentage of chloride (Cl^-) in unknown sample 379 using gravimetric precipitation as AgCl , and report the average result from two independent trials.

Method

The procedure followed Lab 9 of the Chemistry 150 Lab Manual W 2026. Two independent trials were performed.

For each trial, approximately 0.4 g of the unknown was weighed precisely into a 400 mL beaker. 200 mL of deionised water and 5 drops of 6 M HNO₃ were added. 5% (w/v) AgNO₃ solution was added slowly with stirring until the precipitate began to coagulate, then 5 mL more was added. The mixture was heated to near boiling and simmered for a few minutes until the solution cleared. A few drops of AgNO₃ were added to confirm no further precipitation. The beaker was then placed in a dark cupboard for 30 minutes.

A pre-weighed filter paper was prepared. The supernatant was decanted through the filter, the precipitate washed into the filter with small amounts of 0.03 M HNO₃, then rinsed with approximately 15 mL of deionised water. The filter was spread on a watch glass and dried overnight in the oven. The dried filter and precipitate were weighed the following day.

Both trials were run simultaneously — Trial 1 by Dylan Fraser and Trial 2 by David Dreyer. Both beakers were placed in the dark cupboard at the same time and aged for 22 minutes.

Data

Table 1. Raw measurements for both trials.

Measurement	Trial 1	Trial 2
Mass of unknown (g)	0.4160	0.4170
Initial mass of filter paper (g)	0.4690	0.7280
Final mass of filter paper + AgCl (g)	1.4050	1.6630
Mass of AgCl (g)	0.9360	0.9350

Results

Since Ag⁺ and Cl⁻ react in a 1:1 ratio, moles of AgCl equals moles of Cl⁻:

$$n_{\text{Cl}^-} = \frac{m_{\text{AgCl}}}{M_{\text{AgCl}}} = \frac{m_{\text{AgCl}}}{143.32 \text{ g/mol}}$$

The percentage of Cl⁻ in the unknown combines these steps:

$$\% \text{Cl}^- = \frac{m_{\text{AgCl}} \times 35.45}{143.32 \times m_{\text{unknown}}} \times 100\%$$

Trial 1:

$$\% \text{Cl}^- = \frac{0.9360 \times 35.45}{143.32 \times 0.4160} \times 100\% = 55.65\%$$

Trial 2:

$$\% \text{Cl}^- = \frac{0.9350 \times 35.45}{143.32 \times 0.4170} \times 100\% = 55.46\%$$

Table 2. Calculated results for both trials.

	Trial 1	Trial 2
Mass of AgCl (g)	0.9360	0.9350
Moles of Cl ⁻ (mol)	0.006531	0.006524
Mass of Cl ⁻ (g)	0.2315	0.2313
% Cl ⁻	55.65%	55.46%

Average % Cl⁻:

$$\overline{\% \text{Cl}^-} = \frac{55.65 + 55.46}{2} = \boxed{55.56\%}$$

Discussion

Consistency between trials. The two trials gave 55.65% and 55.46%, a difference of only 0.19 percentage points. This is excellent reproducibility for a manual gravimetric procedure and indicates that the precipitation was complete and consistent in both cases.

Ensuring complete precipitation. AgNO₃ was added beyond the point where precipitation stopped and confirmed with a few additional drops. With Ag⁺ in large excess, the K_{sp} expression requires [Cl⁻] to fall to approximately 10⁻⁸ mol/L, meaning the amount of Cl⁻ remaining in solution is negligible.

Coagulation. AgCl initially forms a colloid that passes through filter paper. Heating and the presence of HNO₃ neutralise the surface charges on the colloidal particles, allowing them to coalesce into filterable clumps. The solution clearing during

simmering served as the visual indicator that coagulation was complete.

Light sensitivity. The mixture was kept in a dark cupboard during aging and handled quickly during filtering. Neither precipitate showed significant purple or grey discolouration, suggesting light exposure was not a major source of error in either trial.

Sources of error. The most likely sources of error are incomplete transfer of precipitate from the beaker to the filter (produces a low result) and co-precipitation of AgNO_3 trapped in the colloid during coagulation (produces a high result). Using minimal HNO_3 for washing and avoiding the rubber policeman where possible reduced these effects.

Conclusion

The percentage of Cl^- in unknown sample 379 was determined by precipitating it as AgCl and weighing the dried precipitate. Trial 1 gave 55.65% and Trial 2 gave 55.46%, for an average of **55.56% Cl^-** . The two trials agreed to within 0.19 percentage points, indicating good reproducibility.

Reference

Chemistry 150 Lab Manual W 2026, Lab 9: Gravimetric Analysis of Chloride.